

Department of Mathematical and Physical Sciences
East West University
Course Title: Engineering Physics-II
Course No.: PHY 209
Assignment 2
Full marks: 100
Answer All of the Following Questions

1. Consider the wave function

$$\Psi(x, t) = Ae^{-\lambda|x|}e^{-i\omega t}$$

where A , λ , and ω are positive real constants.

- (a) Normalize Ψ
(b) Determine the expectation values of x and x^2
2. Find the first excited state of the harmonic oscillator.
3. Find the $\langle x \rangle$, $\langle p \rangle$ and $\langle x^2 \rangle$ for the ground state $\psi_0(x)$ of harmonic oscillator.
4. A particle of mass m is bound in the delta function well $V(x) = -\alpha\delta(x)$. What is the probability that a measurement of its momentum would yield a value greater than $p_0 = m\alpha/\hbar$?
5. Find the eigenfunctions and eigenvalues of momentum operator.
6. Prove the following commutator:
(a) $[AB, C] = A[B, C] + [A, C]B$
(b) $[x^n, P] = i\hbar nx^{n-1}$
7. Starting from the Rodrigues formula, derive the orthonormality condition for Legendre polynomials:

$$\int_{-1}^1 P_l(x)P_{l'}(x)dx = \left(\frac{2}{2l+1}\right)\delta_{ll'}$$

8. Prove the following commutator relation:

$$[L_z, x] = i\hbar y, [L_z, y] = -i\hbar x, [L_z, z] = 0$$
$$[L_z, P_x] = i\hbar P_y, [L_z, P_y] = -i\hbar P_x, [L_z, P_z] = 0$$